Solution Architecture

Date	7 July 2026
Team ID	SWTID-2026-3702
Project Name	**A Comprehensive Measure of Well-Being**
Maximum Marks	5 Marks

Solution Architecture Diagram

High-level visual representation of the HDI Predictor's architecture, mapping the Presentation, Application, Prediction, and Data layers.

Presentation / Client Layer

Web Browser — home.html, indexnew.html, resultnew.html

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Application Layer (Flask)

app.py — URL routing, request handling, response rendering

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Input Validation	**Prediction Engine**	**Category Classifier**
request.form parsing	*model.predict(df)*	*Low/Medium/High/V.High*

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Model / Data Storage Layer

model/HDI.pkl + dataset/HDI.csv (195 countries x 82 columns)

SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Component Description Table

Component	Description / Role	Technologies Used
index.html	Landing page displaying HDI introduction, significance, and 'Get Prediction' button	HTML5, CSS3, inline styling
predict.html	Prediction input form with country dropdown and input fields for 4 HDI indicators	HTML5, CSS3, HTML form (POST)
Result.html	Result display page showing predicted HDI score and development category	HTML5, CSS3, Jinja2 template
app.py (Flask)	Core backend handling all routes, model loading, input processing, and prediction logic	Python, Flask, Pickle, NumPy, Pandas
HDI.pkl	Serialized trained Linear Regression model for runtime prediction without retraining	Python Pickle, Scikit-learn
hdi_model.ipynb	Jupyter notebook containing complete ML pipeline: EDA → Preprocessing → Training → Evaluation	Jupyter, Pandas, NumPy, Seaborn, Matplotlib, Scikit-learn
HDI.csv	Source dataset containing 195 countries and 82 human development indicators	CSV format, Kaggle/UNDP source